



March 14, 2025

Mr. Faisal D'Souza
Technical Coordinator
National Coordination Office
National Science Foundation
2415 Eisenhower Avenue
Alexandria, VA 22314

Re: Request for Information (RF) on the Development of an Artificial Intelligence (AI) Action Plan

Dear Mr. D'Souza,

On behalf of the Association of Public and Land-grant Universities (APLU), we appreciate the opportunity to respond to the U.S. National Science Foundation's (NSF) RFI on an AI Action Plan on behalf of the White House Office of Science and Technology Policy (OSTP). APLU's response focuses on the unique role that public research universities play in ensuring American AI dominance by complementing, and working in partnership with, the groundbreaking AI work taking place in private industry.

APLU is a membership organization that fosters a community of university leaders collectively working to advance the mission of public research universities. The association's U.S membership consists of more than 230 public research universities and systems across all 50 states. The association and its members collectively focus on student success and workforce readiness; promoting pathbreaking scientific research; and bolstering economic and community engagement.

The growth of AI has spurred many task forces and pilot programs to determine on how research universities can use these new tools to strengthen the efficiency and security of the research ecosystem, set technical standards, and contribute to further advances in AI that propel that nation's economic competitiveness. APLU convened a group of senior research officers of member universities to discuss this RFI, and they provided several insights grouped into policies for advancing AI energy efficiency, ensuring safe data usage, building trust in the AI ecosystem, encouraging small business, training the next generation of skilled workers through hands on training and advanced infrastructure, and supporting a whole-of-government approach to AI innovation. APLU welcomes the opportunity to partner with the administration in furtherance of shared goals of enhancing U.S. AI competitiveness and furthering economic interests. Public research universities are an optimal partner given unique capacities in scientific research,

longstanding partnerships with industry, and critical role in developing the next generation workforce that holds the promise of U.S. competitiveness in the short and long-term.

Insights to Support American AI Dominance

Co-creating Advances in AI Energy Efficiency

Private industry and academia are united in concern regarding the energy footprint of model training and end-user usage of AI. Evident by Microsoftⁱ and Googleⁱⁱ's exploration of nuclear power use, there is tremendous demand by private industry to expand and strengthen the necessary power systems required by energy intensive AI tools. Advances in both yield and efficiency will enable AI tools to be reliably used in day-to-day life. Universities have a long history of successful research in broadening energy sources through collaboration with agencies such as the U.S. Department of Energy (DOE) in energy programs related to nuclear power and grid resiliency. Public dissemination of results from these advancements enables growth across the marketplace, from small businesses to conglomerates.

Recommendation 1: The AI Action Plan should outline a funding plan for federal programs to support research of energy-efficient AI that benefit the full marketplace, power the development of the next generation of AI tools, and support the Administration's Executive Order "Unleashing American Energy"iii.

Leveraging Robust Data from America's Research Universities to Accelerate Science

The academic sector is primed to address universal AI concerns such as data curation and the use of AI tools in domain specific ways to accelerate American innovation. Public research universities embrace the whole trajectory of AI powered tools, from hardware and data curation to final application. They also host rich portfolios of basic research data spanning astronomy to zoology. The combination of these experiences has created a sector prepared for the time-consuming task of cleaning and curating heterogeneous data sets and identifying new uses for shared tools in domain specific instances. For example, if biomedical researchers can use an AI co-scientist to automate analyses of samples, the researchers can focus on the underlying hypothesis and results of the experiment.

Recommendation 2: As the Administration considers how to accelerate American science, the AI Action Plan should support pilot programs focused on the curation and use of heterogeneous data sets through discipline-specific trials.

Building Trust in AI

Similar to the data curation suggestion above, universities can serve as a third-party validator for open-source data sets for large language models. Rather than dictating any specific data to be used, universities can use their experts to examine commonly used data sets to give them a 'clean bill of health' asserting that the data set has not been tampered with or deviates from its described contents.

Recommendation 3: *The AI Action Plan should support funding plans and encourage research that authenticates underlying data to cultivate trust in the AI tools.*

Encourage Start Ups and Small Businesses Across the Nation

Public research universities provide an entrepreneurial environment where researchers and students can take a breakthrough idea and commercialize the result. APLU supports these activities through a rich history of linking research with economic development, for example through the APLU Innovation & Economic Prosperity Program^{iv}. The Administration should support policies which lower the administrative burden of academic startups and industry partnerships, especially in AI driven inventions. For example, compute credits are a mechanism where an entity is ‘credited’ a specific amount of high-performance computing power or time, at National Labs or potentially supported by industry.

Recommendation 4: *The AI Action Plan should support funding plans for compute credits for small businesses interested in collaborating with academia to connect collaborators in each sector. At a larger scale, the AI Action Plan should support regional- or state-based shared service centers, affirmations of the patentability of new AI innovations, and broad collaboration between the growing number of state-based AI initiatives to share successful practices.*

Training the Workforce for Tomorrow Through Hands-on Training and Infrastructure

Research universities educate the next generation of innovators, programmers, and entrepreneurs. Yet there remains substantial demand for additional science and technology trained workers. The Science & Technology Action Committee (STAC)^v identified labor market gaps facing the U.S. semiconductor industry^{vi}. Additionally, the U.S. National Science Foundation (NSF)’s 2022-2026 Strategic Plan^{vii} highlights the agency’s objective to unleash STEM talent for America.

Recommendation 5: *The AI Action Plan should support programs to train STEM graduates through targeted investment in AI, semiconductor, computer science, and computer engineering related fields across the full spectrum of qualifications (e.g. 2-year, 4-year, graduate levels). The AI Action Plan should also support NSF through recommendations to fund workforce development programs and AI infrastructure at research universities to ensure access to hands-on training across the country.*

Whole-of-Government Support of AI

Federal agencies responded swiftly to recent advances in AI tools; the U.S. National Science Foundation stood up the National Artificial Intelligence Research Resource Pilot, Department of Energy launched the Frontiers in Artificial Intelligence for Science, Security and Technology initiative, and the National Institutes of Health created the Bridge2AI program. These programs have stimulated innovation across the R&D ecosystem through making compute, resources, and models accessible to the nation’s researchers.

Recommendation 6: *The AI Action Plan should build upon the momentum from, and support the funding of, the wide range of AI investments already in progress across the federal agencies.*

Thank you for providing the opportunity for the community to respond. APLU's extensive network of public research universities, our long history supporting basic research and innovation, and our mission to build the workforce of tomorrow underline the above recommendations connect these recommendations to the needs of the AI Action plan, and APLU welcomes the opportunity to discuss our work and recommendations in more detail.

Sincerely,

Deborah Altenburg

Vice President, Research Policy and Advocacy
Association of Public and Land-grant Universities

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ⁱ <https://www.constellationenergy.com/newsroom/2024/Constellation-to-Launch-Crane-Clean-Energy-Center-Restoring-Jobs-and-Carbon-Free-Power-to-The-Grid.html>

ⁱⁱ <https://blog.google/outreach-initiatives/sustainability/google-kairos-power-nuclear-energy-agreement/>

ⁱⁱⁱ <https://www.whitehouse.gov/presidential-actions/2025/01/unleashing-american-energy/>

^{iv} <https://www.aplu.org/our-work/innovation-and-economic-prosperity-universities-designation-and-awards-program/>

^v <https://sciencetechaction.org/committee>

^{vi} <https://www.semiconductors.org/chipping-away-assessing-and-addressing-the-labor-market-gap-facing-the-u-s-semiconductor-industry/>

^{vii} <https://www.nsf.gov/reports/planning/leading-world-discovery-innovation-stem-talent-development>